

MATH 4323 Project Part 4: Final report

Please include your name and student id in your final report. If you decide to work as a group, you will have to include the names and student ids of all group members.

Submission remarks:

- The text in your project should be in 12 point Times New Roman font. Your paragraphs should use single line spacing. Use 1 inch top, bottom, right, and left margins. Length: 6 -10 pages. All contents beyond 10 pages should go into the appendix.
- Provide only the most critical pieces of code in this report, without including too much output and clogging up the report. Simply attach the full R source code and your data set to your submission in order for me to have access to the whole thing.
- If you decided to work as a group, for each section/subsection, make sure to specify the names of students who mostly worked on that part of the report. E.g. "2. Hierarchical Clustering (Johnny Doey, Brenda Parker)".

Detailed Requirements:

1. **Introduction:** You could provide some background knowledge about the data you are analyzing, clearly and concisely present your research question (one main overall question, which deals with one response variable; try to avoid asking multiple questions, each about a different response variable), describe the data set that can help answer your question; 0.5 -1 page.

2. **Methodology:** In this section, you could describe the methods (at least two distinct models covered in this course) you used in your analysis, along side the reasons why, and what are the expected advantages/disadvantages of each method. You will need to write down the model formula: If it is KNN, write how we calculate the predictions for the observation based on its neighbors; If it is SVM - write the hyperplane equations corresponding to selected kernels; If it is K-Means clustering, write the explicit formula for main optimization criteria; If it is hierarchical clustering, explain how clusters are merged depending on the linkage used. This section can be 0.5 to 1 page.

3. **Data Analysis:** In this section, you should address the questions of interest and interpret the results in terms of the questions of interest you proposed, including relevant tables and figures (please adjust your figures to appropriate sizes); 1-5 pages.

For this section,

- (a) You need to walk me through the thought process of your considerations while

fitting the model. E.g. would you consider excluding some predictors from consideration, and why? Some predictors might not make sense from domain knowledge, or they might be highly correlated with other predictors. What about scaling the data - is it necessary, and why?

(b) Select the optimal tuning parameter value for your model. For example, select optimal K for KNN or K-Means; select optimal values of cost, γ for your SVM; select optimal linkage for your hierarchical clustering (by selecting optimal K for each linkage beforehand).

If you are performing supervised learning (KNN, SVM), then

- i. Randomly subdivide your full data set in 80% for training, and 20% for testing (Note: make sure to use `set.seed()` in order for both models to work on the train/test data subdivision.)
- ii. Proceed to train your model on the 80% training data, and then record its prediction error on left out 20% testing data.

If you are performing unsupervised learning (clustering), then

- i. Make sure to calculate the silhouette coefficient for the optimal clustering.

(c) For supervised learning: compare the prediction test error resulting from part (b) for both models; For unsupervised learning: compare optimal silhouette coefficients for both clustering methods; use external class labels (if available) to check if the clustering makes sense.

(d) Fit your **best** model from part (c) on the full data set (no train/test subdivision), and output the results, providing the most important model summaries and images (if applicable) that resulted from your model fitting. For KNN – summary of your KNN object, its classification performance on full data. For SVM - summary of SVM object, of support vectors in each class, a plot of your fitted boundary with respect to some pairs of predictors (see one of HW assignments).

For K-means clustering - use `eclust()` with its 2D plot; for hierarchical clustering - output dendrogram.

(e) Provide the interpretation of results and your conclusions as it pertains to the original overall question. E.g. for supervised learning, did you achieve a respectable predictive performance? For unsupervised, what kind of groups you've obtained? Do they make sense?

4. **Conclusion:** Answer the questions in the introduction and comment on the difficulties you have in the data analysis. Some possible procedures you think that can improve the data analysis.

5. **References:** List the references that are cited in your project.

6. **Appendix (optional):** Include all supporting materials, for example, additional figures/tables, R code, technical derivations.